Advanced Condition Monitoring for Active Front-End Converter LCL Filter AC Capacitors

Edi Matijevic, *Member, IEEE*, Rahul Sharma, *Senior Member, IEEE*, Ajay Hemanth, Firuz Zare, *Fellow, IEEE*, and Dinesh Kumar, *Senior Member, IEEE*

***Abstract*— The increasing penetration of power electronics in recent times has resulted in a higher level of harmonics, leading to accelerated degradation of filter capacitors used in power electronics converters. As a result of the increased degradation caused by harmonics, capacitor condition monitoring has gained significant relevance as a means to reduce failures and subsequently lower maintenance costs. While state-of-the-art research primarily emphasizes dc-link capacitor monitoring, industry reports indicate that ac capacitors in LCL filters for motor drive applications are prone to accelerated failures. This article proposes a control method and algorithm to address the challenge of monitoring ac capacitors within the LCL filter employed in power electronics converters. This article presents a novel condition monitoring control method that is based on the collective use of artificial neural network (ANN) and extremum-seeking techniques to deliver prompt and accurate estimation of the capacitance of ac capacitors in LCL filters. The effectiveness of the condition monitoring control method has been validated through comprehensive simulations and further verified through experimental testing. The results demonstrate that the proposed condition monitoring control is highly accurate, fast, and computationally efficient.**

***Index Terms*— AC film capacitor condition monitoring, active front-end(AFE) converter, motor drives.**

I. Introduction

ACTIVE front-end (AFE) converters are crucial in motor drive applications, but they introduce high-frequency switching, requiring filters to remove harmonics. The LCL filter is a popular choice for this purpose. However, the introduction of an LCL filter reduces the reliability of AFE converters, as the capacitor is one of the first components to fail in a power electronic systems [1]. Furthermore, the degradation of the capacitor in the LCL filter can alter the filter's resonant frequency, which, in turn, can impact the stability of the entire system [2]. Hence, monitoring the capacitor's condition offers a potential solution to this problem.

Condition monitoring has become a well-established field in the last decade. Typically, the deterioration of a capacitor can be ascertained by observing specific parameters that indicate this degradation. Mainly, the parameters for estimating the degradation of capacitance can be divided into nonelectrical and electrical parameters. Nonelectrical parameters involve measuring attributes, such as weight, internal pressure, internal temperature, or structural integrity, to estimate capacitor degradation. In [3], an acoustics detection method utilizes the acoustic response of MLC-Caps to distinguish between damaged and undamaged capacitors using a support vector machine (SVM) classifier. However, this method as the other methods based on nonelectrical parameters requires additional sensors with high accuracy, which increases the cost. Another challenge with this type of condition monitoring is the lack of uniform criteria for determining the end of life. This makes it not suitable for condition monitoring of ac capacitors in LCL filter converters.

On the other hand, measuring electrical parameters, such as dissipation factor, capacitor impedance, equivalent series resistance (ESR), and capacitance, can provide insights into capacitor degradation. Electrical parameters, such as dissipation factor and capacitor impedance, are hard to measure and use to define end of the life, so the majority of condition-monitoring methods are based on the estimation of capacitance C or ESR. If the measured degradation parameter C falls below a specific threshold or if there is a significant increase in ESR, it is indicative that the capacitor has reached the end of its lifespan. To estimate capacitance or ESR, there are two predominant principles. One involves parameter estimation using the periodic ripple inherent in capacitors, while the other principle relies on nonperiodic large signals occurring during charging or discharging. Sundararajan et al. [4] propose a method based on the first principle, based on the online estimation of the capacitor's ESR using the switching frequency components of the dc-link capacitor voltage and current. Similarly, Hasegawa et al. [5] extract both ESR and capacitance without injecting any additional frequency component. On the other hand, Li et al. [6] utilize signal injection into the dc link for real-time capacitance estimation. Pu et al. [7] induce an ac signal in a dc-link capacitor and use a recursive least squares algorithm for reliable estimation results. In [8],

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Edi Matijevic, Rahul Sharma, and Ajay Hemanth are with the School of Electrical Engineering and Computer Science, The University Of Queensland, Brisbane, QLD 4072, Australia (e-mail: matijevic.edi@gmail.com).

Firuz Zare is with the School of Electrical Engineering and Robotics, Queensland University of Technology, Brisbane, QLD 4000, Australia.

Dinesh Kumar is with the Department of Power Electronics Engineering, Danfoss Drives A/S, 6300 Gråsten, Denmark.

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degradation indicators ESR and capacitance are estimated from dc-link voltage and stator current measurements, while in [9], the estimation of capacitance and ESR is achieved by leveraging the magnitude and phase shift characteristics of the LCR network inherently present in variable speed drives.

Ghadrdan et al. [10] use the second principle and focus on the condition monitoring of dc-link capacitors in three-phase back-to-back converters using measurable transients, achieving accurate capacitance estimation with less than 1% error. Bărbulescu et al. [11] design a parameters observer for estimating capacitance and ESR using the discharge profile of the capacitor. Sundararajan et al. [12], leverage not only capacitance and ESR data but also temperature information to accurately determine capacitor degradation. In addition, there are data-driven condition monitoring techniques; for instance, researchers in [13] employ artificial neural networks (ANNs) in dc-link capacitors to estimate capacitance using current and voltage data. In general, most of the condition monitoring techniques are based on the same principle, and for more information on different techniques, the reader can refer to the papers in [14] and [15] that present a comprehensive overview of different condition-monitoring techniques for dc-link applications.

Nevertheless, these studies predominantly concentrate on dc-link capacitors condition monitoring, leaving ac side capacitors largely unaddressed, despite industry reports indicating that they too are prone to failure. Although certain techniques for condition monitoring dc-link capacitors may be applicable to ac capacitors for LCL filter applications, there are specific drawbacks and limitations inherent to their translation. For instance, estimating ac capacitor capacitance using discharge/charge profiles is impractical for LCL filter converter applications. Conversely, employing a periodic ripple-based method necessitates at least one current or voltage sensor, which is suboptimal, as these sensors are typically not directly placed on capacitors in LCL filter converters. To the best of the author's knowledge, at present, only a single study has presented condition monitoring specifically for ac-side capacitors [16]. The authors of this article propose an algorithm that detects if there is a change in resonant frequency and subsequentially drop in capacitance by injecting a control signal; however, they do not consider the effect of grid impedance on a resonant frequency. The impacts of grid impedance on resonant frequency could potentially give rise to false indications of capacitor degradation. Hence, this article introduces a condition monitoring algorithm capable of identifying capacitance degradation without the need for additional sensors, regardless of whether the system is connected to a weak or stiff grid specifically designed for ac capacitors in LCL filter converters.

To precisely estimate capacitor degradation, this article introduces an algorithm that effectively combines ANN and extremum-seeking (ES) control methodologies to yield accurate results. The ANN is used to estimate inductive grid impedance based on available current and voltage measurements, while ES estimates resonant frequency, which then when combined with estimated grid impedance can result in accurate capacitance estimation. The technique outlined in this article represents an online method and can be implemented at any given time, irrespective of grid conditions. This article introduces several novel contributions and advancements in the field, including the following.

1) A novel algorithm for condition monitoring of ac capacitors in AFE systems is presented. The proposed approach combines ANN and ES control approach for enhanced prompt and accurate estimation with minimal computational requirements and with reliance only on the existing measurements.
2) Using voltage and current harmonics as the only input parameters, the approach is based on the sequential estimation of grid impedance through ANN followed by estimation of capacitance degradation using ES, whereby the estimate of grid impedance is injected as a known parameter. In addition to real-time monitoring, another key feature of the proposed approach is that it incorporates a notification system to alert users when a capacitor is nearing the end of its operational life. This proactive approach facilitates timely maintenance and replacement actions. The proposed approach has the potential to revolutionize maintenance practices by promoting a proactive and predictive maintenance approach as opposed to costly corrective approaches, thereby minimizing the risk of unexpected failures.
3) The proposed method is extensively validated through a combination of simulation and experimental studies, which demonstrate that the estimation can be completed in as low as just 0.3 s. These tests highlight its potential for practical implementation in real-world applications.

The subsequent five sections, which form the entirety of this article, are concisely outlined as follows. Section I serves as a comprehensive review of the current state of the art in ac capacitor condition monitoring. Specifically, it establishes the motivation for the research and explains research gaps in the field. In Section II, this article delves into an in-depth problem formulation. This section highlights the necessity of employing ES and ANN techniques for condition monitoring of ac LCL filter capacitors. Section III elucidates the theory behind the proposed condition monitoring algorithm, providing an explanation of extremum-seeking resonance estimation, and ANNs, and presenting a detailed mathematical framework for the proposed algorithm. Section IV conducts comprehensive simulations to validate the ANN grid impedance estimation under diverse grid conditions. In addition, it simulates the entire condition-monitoring algorithm to assess its performance. Section V offers experimental validation for the proposed ac capacitor condition monitoring algorithm implemented in an AFE setup. Finally, Section VI summarizes the findings presented in this article.

II. Problem Formulation

In this article, the main focus is on the degradation of ac capacitors used in LCL filters for AFE applications. To counter this inherent ac capacitor characteristic, a condition monitoring approach is introduced. The system of interest, the AFE converter, has voltage sensors already present for phase-locked loop implementation, while the current sensors are typically placed on the grid side or the converter side. This article considers a system where both current and voltage sensors are placed on the grid side at the point of common coupling (PCC).

As discussed in the introduction, ongoing research on capacitor condition monitoring mainly focuses on dc-link capacitors; however, ac capacitors also experience temperature variation due to harmonics leading to premature failures. AC capacitors primarily function as filters to mitigate harmonic components, as they provide a low-impedance path for high-frequency currents for both, harmonics produced by nonlinear load and harmonics originating from PCC. Despite selecting capacitors suitable for specific applications, the unpredictability of harmonics at the PCC can result in capacitors experiencing higher-than-expected temperature stress and accelerated degradation.

Metalized polypropylene film (MPPF) capacitors are commonly employed in ac LCL filter applications due to their well-suited characteristics for ac purposes, such as high-voltage ratings, low ESR, good frequency characteristics, and high-current-carrying capability. The high-temperature stress induced by harmonics can result in the breakdown of the dielectric and the formation of defects in the capacitor's dielectric layers. Just as electrolytic capacitors, MPPF capacitors have the self-healing ability, repairing defects in the dielectric layer by anodic oxidation [17]. These burned defects, although isolated, contribute to a small loss of dielectric film, leading to a minor decrease in capacitance. Over time, the capacitance significantly decreases, giving rise to various issues and potentially causing complete failures.

In addition to the possibility of complete failure, capacitance degradation has significant impacts on the overall system performance. First and foremost, capacitance degradation results in a decrease in the overall attenuation of the LCL filter and an increase in THD. Furthermore, with the reduction in filtering, the capacitor experiences an increase in temperature stress due to an increase in harmonics, thus accelerating the failure process [18]. In addition to that, a change in LCL filter resonant frequency could have a serious impact on the stability and damping of the system. Therefore, considering all the above, monitoring the capacitor and replacing it before the system experiences a drastic drop in performance are important.

The primary objective of implementing condition monitoring is the prevention of capacitor failure and performance loss. Furthermore, proper maintenance and replacement before failure drastically reduce cost and increase the life hours of the whole system.

In this article, the resonant frequency of the LCL filter serves as a crucial attribute to effectively monitor the capacitance of the LCL filter capacitor. When capacitance changes, so does the resonant frequency of the LCL filter. Therefore, by measuring the resonant frequency of the LCL filter, the capacitor can be monitored. To achieve accurate results, it is crucial to identify the impact of grid impedance on the resonant frequency. In this consideration, the assumption is made that the grid is inductive, and only the effect of grid inductance

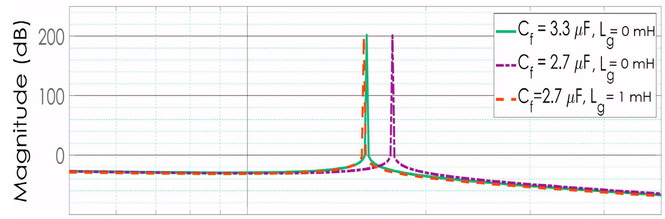

Fig. 1. Bode diagram showcasing the effects of capacitor degradation and grid impedance on AFE LCL filter resonant frequency.

is taken into account, as the impact of grid resistance on the resonant frequency is deemed minimal. Fig. 1 showcases the effect of capacitor degradation and grid impedance on an undamped AFE LCL filter converter.

When the capacitor value in the LCL filter is lowered by 20% (3.3 $\rightarrow$ 2.7 μF), the resonant frequency increases, but if grid impedance is 1 mH, the resonant frequency increase is negated. The equation that describes this behavior is the following:

$$\omega_{\text{res}} = \sqrt{\frac{L_1 + L_2 + L_g}{C_f L_1 (L_2 + L_g)}} \tag{1}$$

where L_1 and L_2 are converter and grid side inductors, ω_{res} is the LCL filter resonant frequency, and C_f is a filter capacitor, while L_g is grid inductive impedance. This equation and figure clearly demonstrate that capacitor degradation leads to an elevation in the resonant frequency, whereas an increase in grid impedance contributes to a lower resonant frequency. Hence, while it is possible to estimate the resonant frequency, the determination of the capacitor's end-of-life is not feasible without knowledge of the grid impedance. In fact, (1) is the backbone of the proposed algorithm for online monitoring of LCL filter capacitors. The equation demonstrates that given the knowledge of other variables, it is feasible to estimate the capacitance C_f. This concept is explored in greater detail throughout the remaining sections of this article.

Section III provide an in-depth explanation of how the resonant frequency of the LCL filter can be used to estimate capacitance and how grid impedance can be estimated from voltage and current harmonics.

III. Condition Monitoring Algorithm

As detailed in Section II, the process for estimating the capacitance of ac capacitors within the LCL filter involves two main steps. Initially, ES is utilized to determine the resonant frequency of the LCL filter, after which ANN is employed to estimate the grid impedance. Once both estimates are acquired, the capacitance can be accurately determined. The proposed LCL filter ac capacitor condition monitoring control is shown in Fig. 2. The figure provides an overview of the condition monitoring control system, illustrating the key components and their interconnections. Sections III-A–III-D delves into a comprehensive examination of each individual block, providing a detailed explanation of their functions and operation. As seen in the figure, the AFE is connected to the grid through the LCL filter; voltage and current are measured at the PCC through sensors, and relevant data are extracted. Using the sampled data, the magnitudes of the fifth harmonic are extracted from the voltage and current signals. The extracted harmonics are subsequently utilized to calculate the reactance component (X), and these three variables are employed by the ANN to estimate the grid impedance. At the same time, ES control injects variable frequency signal U_{inj} on top of the AFE control signal U_{ref} and uses the grid current measurements to estimate the resonant frequency, consequently combining an estimated grid impedance $L_g$est and estimated resonant frequency ω_{res}est to monitor and estimate capacitor capacitance. The entire condition monitoring algorithm, including both ANN and ES, operates at the sampling frequency of the measurements and the controller. This frequency is consistent with the switching frequency of the converter, set at 10 kHz for this project. Should the estimated capacitance drop below 90% of its initial value, the control system indicates the need for maintenance. While the end of the lifetime for MPPF ac capacitors is usually specified by the manufacturer in the datasheet, the rationale behind selecting a 10% variation from the initial capacitance value as the end of the lifetime is based on practical considerations. A 10% threshold strikes a balance: it signifies a significant enough deviation to potentially affect the performance of converters, yet it avoids excessive costs associated with frequent capacitor replacements if the end of the lifetime threshold was set lower. However, if a lower threshold for the end of the lifetime is deemed necessary, it can be accommodated within the algorithm without any issues. This whole process is extremely fast and takes under 0.3 s when initiated. Given the assumption that the three phases are predominantly balanced and capacitors encounter similar amounts of harmonics and temperature stress, monitoring only one capacitor is sufficient, as they are expected to degrade at approximately the same rate. Hence, the algorithm utilizes only voltage and current measurements from phase A for monitoring the ac capacitor in that specific phase under that assumption. In cases where the assumption of predominantly balanced phases may not hold true, there is a simple solution to monitor all three ac capacitors. Instead of executing the algorithm once to estimate capacitance in phase A, it is run sequentially for all three phases. This approach allows for the monitoring of all three capacitors, as a trade-off of consuming more computational power for a short duration. It is also important to mention that this is the top overview of the proposed control, and Sections III-A–III-D will offer deeper insight into the functionality of each block. Furthermore, an algorithm flowchart encompassing all the processes will be provided to facilitate a clearer understanding of the control system's operation.

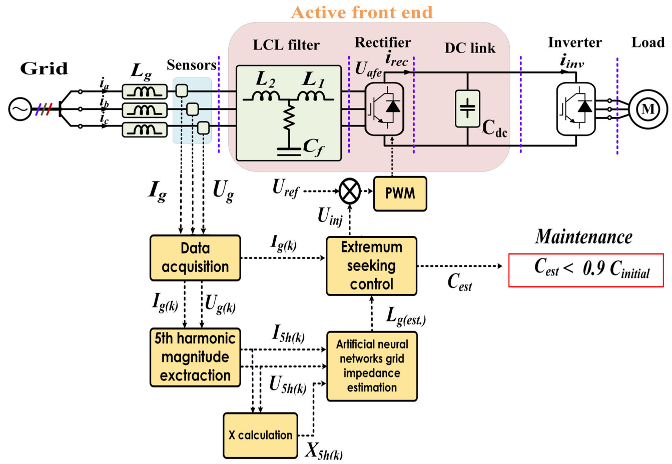

Fig. 2. Top overview of the proposed condition monitoring control.

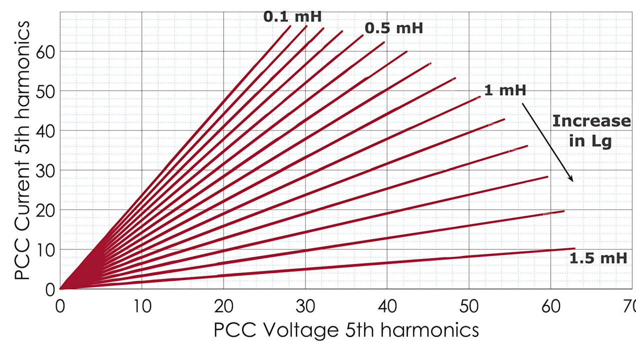

Fig. 3. PCC voltage and current harmonics magnitude under grid impedance variation.

A. ANN-Based Grid Impedance Estimation

To illustrate how an ANN can estimate grid impedance using voltage and current harmonics, it is crucial to delve into the relationship between these elements. To highlight this relationship, a nonlinear load was simulated under a range of load conditions and variations in grid inductance.

Fig. 3 shows that grid inductance has a nonlinear effect on voltage and current harmonics. Here, it can be observed that the rate of change of gradient is nonlinear for a linear change in grid inductance. As the impedance of the grid increases, the magnitude of current harmonics decreases, while the magnitude of voltage harmonics increases. This creates a nonlinear relationship among voltage harmonics, current harmonic, and grid inductance. This article exploits this nonlinear relationship between current and voltage harmonic to identify the grid inductance. ANNs are well suited for addressing this regression problem, as they are capable of capturing and quantifying the underlying pattern in the data.

Due to fifth-order harmonics being dominant harmonics in three-phase motor drives, this article uses fifth-order current and voltage harmonics to identify the grid inductance L_g.

Fig. 4 shows a structure of ANN used for grid impedance estimation. As shown in the figure, the grid impedance estimation starts with three input neurons. The selection of input neurons comes from Fig. 3, which illustrates the complex interplay among voltage, current harmonics, and grid impedance. This intricate relationship suggests that an ideal choice for the neural network inputs would be the voltage fifth harmonics and current fifth harmonics, represented by two input neurons. To increase the accuracy, a third neuron is added as reactance X as seen in the following equation:

$$X_{5h} = \frac{V_{5h}}{2\,\pi\ 250\ I_{I_{5h}}}. \tag{2}$$

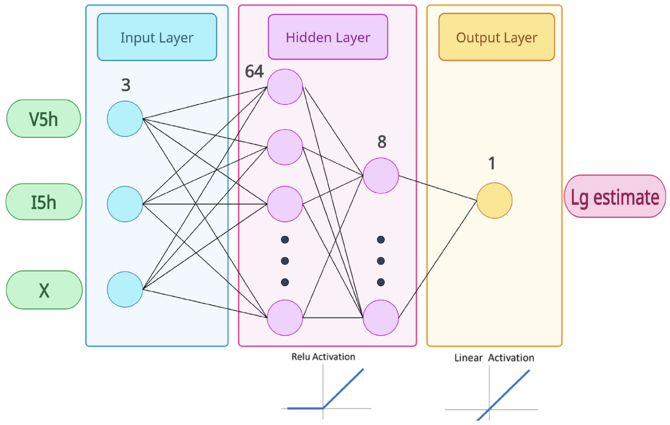

Fig. 4. ANN grid impedance estimation block.

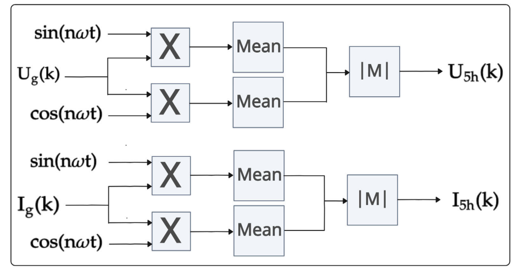

Fig. 5. Visualization of the fifth-harmonic magnitude extraction.

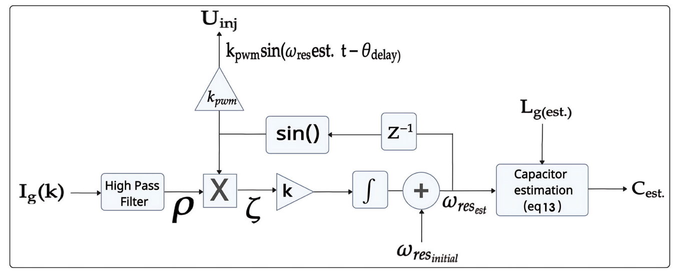

Fig. 6. ES control block with capacitor estimation.

The output layer comprises a single neuron dedicated to estimating the grid impedance. The hidden layers of the neural network architecture comprise two layers, with 64 neurons in the first layer and 8 neurons in the second layer. For this particular problem, hidden layers use ReLU activation (3) function due to its characteristics to capture nonlinear patterns, while the output activation function uses linear activation (4)

$$\mathrm{ReLU}(x) = \begin{cases} 0, & \text{if } x < 0 \\ X, & \text{if } x > 0 \end{cases} \tag{3}$$

$$\mathrm{Lin}(x) = x. \tag{4}$$

The input node gets the data, the hidden layer processes the data, and the output layer produces the output prediction. Each section has a "neuron" (represented by a circle) where data are received from the previous section, and mathematical operation is performed on the data. Each neuron is connected to all the neurons in the consecutive section through weights. During the training process, the weights and biases of the model are adjusted to effectively capture the intricate patterns and correlations required for accurate prediction.

The training data for the ANN were collected from simulations conducted on an AFE converter system in SIMULINK, encompassing diverse load scenarios and variations in grid impedance. Sparing unnecessary details, the results and accuracy of the trained model are explored in Section IV.

B. Fifth-Harmonic Magnitude Extraction

As stated earlier in the preceding subsection, in order for the ANN to accurately estimate the grid impedance, it requires the magnitudes of specific harmonics for both the current and voltage signals. Due to the specific focus on AFE and motor drives, particular emphasis is placed on the magnitude of the fifth harmonics, as they are predominant in these types of systems. Fifth-harmonic magnitudes are calculated using the Fourier series, as shown in Fig. 5. The measured samples of voltage and current are multiplied by both sine and cosine, and the mean is calculated. The mean value is calculated by averaging over a sliding window of one cycle duration based on the specified fundamental frequency. Magnitudes are extracted from resulting Fourier series coefficients.

C. ES-Based Resonant Frequency Estimation

To estimate the resonant frequency of the LCL filter, extremum-seeking control is used. ES is an optimization method capable of finding the local extremum only based on the output measurements without knowing the plant model. Therefore, an application to estimate the extremum of the LCL filter, the resonant frequency is a perfect task for ES control.

ES control, originally applied in various engineering fields [19], has more recently found applications in power electronic. Matijevic et al. [20] and Wu et al. [21] utilized ES techniques to estimate the resonant frequency for the purpose of implementing adaptive active damping control without injection. On the other hand, Wu et al. [22] use injection-based ES to estimate resonant frequency. To date, there has been no research paper that applies ES control specifically for the purpose of capacitor condition monitoring. Thus, Section III-C dives into an explanation of how ES is used for condition monitoring purposes.

This article utilizes the characteristic of the LCL filter, specifically the low impedance observed at the resonant frequency, as depicted in Fig. 1. ES control introduces a variable high-frequency signal that modifies the frequency, aiming to reach the extremum when $\omega_{\text{res}}\text{est.} = \omega_{\text{res}}$. In other words, ES control converges when it identifies the resonant frequency of the LCL filter. Therefore, by injecting a variable frequency signal, the ES control can probe and determine a direction that drives ES control to its maximum/extremum point, LCL filter resonant frequency.

Fig. 6 shows the ES control implementation followed by a mathematical explanation of the system for this application. The objective is to find the extremum, resonance point of the LCL filter, as seen in Fig. 1. Assuming that the plant has reached a steady state, the plant can be viewed as a static map, where output y is the function of U_{inj} and U_{inj} is a function of estimated resonant frequency $\omega_{\text{res}}\text{est}$. Thus, we have the following y output and J as a function to be maximized:

$$y = J\left(U_{\text{inj}}\right) = J\left(k_{\text{pwm}} \sin\left(\omega_{\text{res}}\text{est}\, t \;+\; \theta_{\text{delay}}\right)\right). \tag{5}$$

The probing signal is itself a function of injected frequency ω_{res}est that at the end determines whether the current magnitude will increase or decrease. Knowing that the LCL filter has a resonant frequency, thus, extremum, then optimal U^*_{inj} exists at the corresponding frequency $\omega_{\text{res}} = \omega_{\text{res}}$est. Measured and sampled grid current $I_g(k)$ mainly contains three different parts, such as fundamental frequency, other harmonics and, most importantly, injected frequency current, and a gradient of current magnitude β with injected probing voltage U_{inj}

$$I_g(k) = \underbrace{A_1\;\sin(\;\omega_1 t)}_{\text{Fundamental}} + \underbrace{\beta\;\sin(\;\omega_{\text{res}}\text{est.}\;t)}_{\text{Injected frequency}}\;\underbrace{\frac{\partial\beta}{\partial U_{\text{inj}}}}_{\text{Gradient}} \;+\; \underbrace{X(t)}_{\text{Other harmonics}}\;. \tag{6}$$

The measured current $I_g(k)$ from (6) is then processed through a high pass leading to a derivation (7). A high-pass filter removes fundamental current and low-order harmonics. After passing through the high-pass filter, only the injected frequency current and certain higher harmonics remain. However, to simplify the analysis and because the injected frequency current has a higher magnitude, other harmonics are disregarded

$$\begin{aligned}\rho &= \text{HPF}\left[I_g(k)\right]\\ \rho &\approx A\beta\;\sin(\;\omega_{\text{res}t}.\text{est}\;t)\;\frac{\partial\beta}{\partial U_{\text{inj}}}\end{aligned} \tag{7}$$

where A is the attenuation factor due to the high-pass filter. High-pass filter cutoff frequency is chosen to be smaller than estimated resonance ω_{res}est but large enough to remove fundamental frequency

$$\begin{aligned}\zeta \;&=\; \sin\left(\omega_{\text{res}}\text{est}\,t \;+\; \theta_{\text{delay}}\right)\rho\\ &= A\beta\;\sin(\omega_{\text{res}}\text{est}\,t)\;\;\sin\left(\;\omega_{\text{res}t}.\text{est}\,t \;+\; \theta_{\text{delay}}\right)\frac{\partial\beta}{\partial U_{\text{inj}}}\;.\end{aligned} \tag{8}$$

The filtered signal is then demodulated by multiplying it with the sine of an initial estimate for the resonant frequency. Given that the high-pass filter satisfies this condition

$$\omega_{\text{line}} < 2\omega_h < \omega_{\text{res}} \tag{9}$$

where ω_h cutoff frequency is big enough to remove the fundamental grid frequency ω_{line} but small enough to pass the resonant frequency of the system ω_{res}; then, the product of two sines is positive most of the times. It is evident that ζ serves as an approximation of the gradient, disregarding all the time-varying components. Consequently, by employing gradient-seeking, $U_{\text{inj}}(\omega_{\text{res}}\text{est})$ can be determined that maps to an extremum point, resonant frequency. The integrator then updates ω_{res}est until the extremum of the function is reached. This can be seen by considering discrete zero-hold integrator

$$I_{\tau,k(z)} = \frac{\tau k}{z-1} \tag{10}$$

where τ, k is the sampling time and integrator gain. The k is adjusted to be positive if seeking for a maximum or negative when seeking for a minimum extremum. By applying this integrator to the value of ζ at iteration i (ζ_i), we obtain an update for the estimated parameter ω_{res}est

$$\omega_{\text{res}}\text{est.} = \frac{\tau k}{z-1}\zeta_i + \omega_{\text{res}}\text{initial} \tag{11}$$

$$\omega_{\text{res}}\text{est}_{i+1} = \omega_{\text{res}}\text{est} + \tau k \zeta_i. \tag{12}$$

The initial resonance guess ω_{res}initial is added on top of the estimate to accelerate the process of convergence. Knowing that the $\tau > 0$ and $\zeta > 0$, the term $\tau k\zeta$ sign is only depended on k. The sine signal with the estimated frequency is delayed by one sample to account for internal delays and then injected into the PWM as a probing mechanism. The term $\tau k\zeta$ guides estimate ω_{res}est toward an extremum of the function and, in other words, guides it to the system resonance. Eventually, the extremum is reached when the gradient ($\partial\beta/\partial U_{\text{inj}}$) reaches 0, and the estimate will oscillate around the extremum (due to injected sinusoidal perturbation).

After proving that the convergence can be achieved, the estimated resonant frequency ω_{res}est is used in combination with estimated grid impedance $L_g$est to estimate capacitance using (13) derived from (1)

$$C_{\text{est}} = \frac{L_1 + L_2 + L_g\text{est}}{\omega_{\text{res}}\text{est}^2\;L_1\left(L_2 + L_g\text{est}\right)}. \tag{13}$$

Hence, by determining the LCL filter resonant frequency and grid impedance, it becomes possible to estimate the value of the filter capacitor, assuming the known inductance values of the LCL filter.

D. Algorithm

The algorithm depicted in Fig. 7 outlines the complete condition monitoring process. Since the capacitor degradation is a slow-varying process, the algorithm can be run once per day, ideally at similar times and conditions to eliminate external variables.

The entire process has a duration of 0.3 s and begins with the acquisition of measurements for grid current, grid voltage, and dc-link voltage. Following data acquisition, ES is activated for the duration of 0.2 s, and upon its deactivation, the final value of resonance estimation ω_{res}est is recorded and stored. A precautionary step is implemented before activating the ANN for grid impedance estimation, involving the verification of system stability in a steady-state condition. In AFE converters, the dc link is controlled, and when there is a transient state, the dc-link voltage will differ from dc-link voltage reference U^*_{dc}. Therefore, if the difference between real dc-link voltage and dc-link voltage reference is below 50 V, for the system controlled at 600 V, this is considered a steady state. Certainly, depending on the specific application, it is possible to reduce the differences. Taking all factors into consideration, this approach aids in the faster convergence of the ANN and increases its overall accuracy. At steady state, ANN is enabled to estimate grid impedance $L_g$est for about 0.1 s, after which is deactivated and the last grid impedance value is stored. Using both estimated resonance ω_{res}est and estimated grid impedance $L_g$est, filter capacitor capacitance

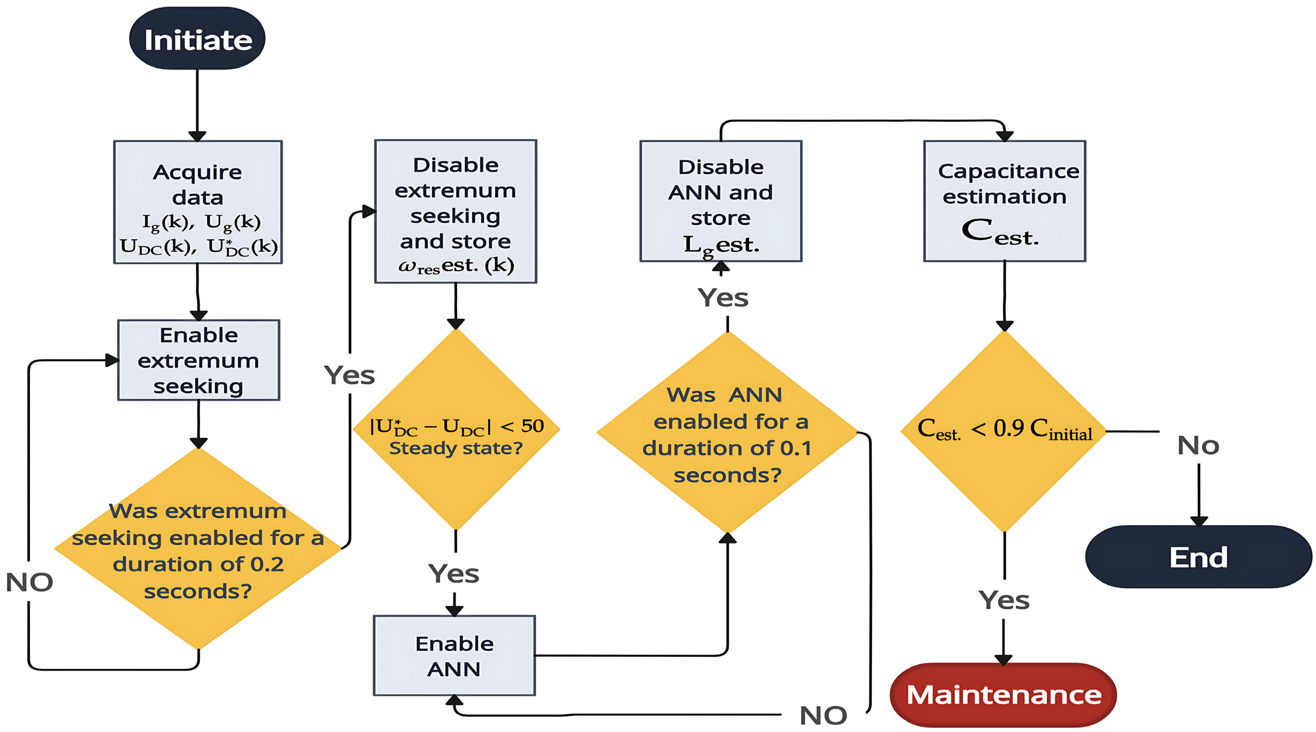

Fig. 7. Condition monitoring algorithm flowchart.

is calculated using (13). In the final step of the algorithm, the estimated capacitance (13) is compared with 90% of the initial capacitance C_{initial} to determine whether maintenance is necessary or not. It is crucial to note that the initial capacitance of the ac capacitor should be determined prior to enabling the condition monitoring algorithm. The initial capacitance is determined either through manual measurement prior to the commissioning of the AFE drives or by utilizing this algorithm during the initial deployment.

IV. Simulation

Section IV presents a simulation aimed at verifying the effectiveness of both the ANN-based grid impedance estimation and the overall control system for capacitor condition monitoring.

A. ANN Grid Impedance Estimation

To ensure that capacitor estimation is accurate, it is important for grid impedance estimation to be precise under different grid conditions. The performance of the ANN is evaluated under diverse grid conditions, including voltage swell, voltage sag, step changes in grid impedance, and variations in load.

In Fig. 8, the grid impedance estimation by the ANN is represented by the blue curve, while the orange and purple curves represent the magnitudes of voltage and current, respectively. The simulation starts with normal voltage conditions and 1 mH of grid impedance. As depicted in the figure, when we disregard the noise and oscillations, the estimated grid impedance closely aligns with the actual grid impedance of 1 mH. At 0.25 s, when the grid impedance is reduced to 0.5 mH, the ANN successfully tracks the step change with

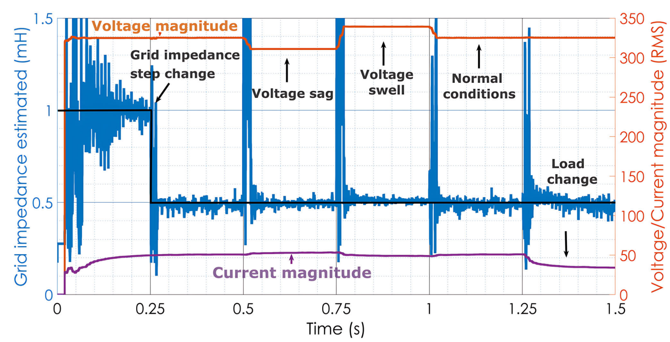

Fig. 8. Simulation of grid impedance estimation under different grid conditions.

high accuracy. At 0.5 s, a voltage sag is observed as the voltage magnitude drops. However, despite the initial transient error, the ANN consistently and accurately estimates the grid impedance. It is important to highlight that the transient error witnessed during the voltage sag is linked to the magnitude extraction block, not the ANN itself. Fortunately, this error does not impact the condition monitoring algorithm, since the algorithm disregards the initial measurement, as it will be shown in Section IV-B. Therefore, if the ANN is disabled during transients, the occurrence of such errors can be mitigated. At 0.75 s, the grid condition changes from voltage sag to voltage swell, and after initial transients, ANN continues to estimate grid impedance accurately. At 1 s, the grid voltage is back to normal, and a load change happens at 1.25 s. As seen in the figure, ANN prediction and estimation are accurate even under different grid voltage conditions. In conclusion, the simulation results demonstrate the robustness of

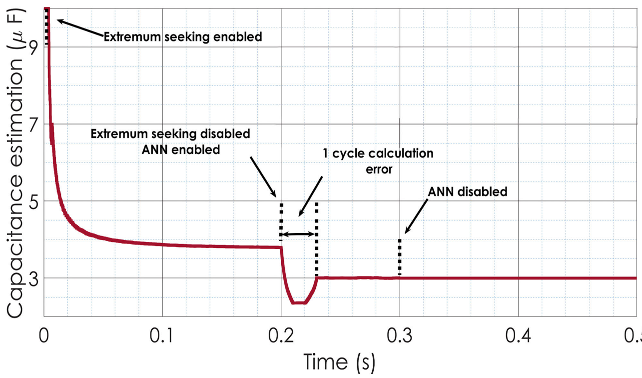

Fig. 9. Condition monitoring algorithm capacitance estimation.

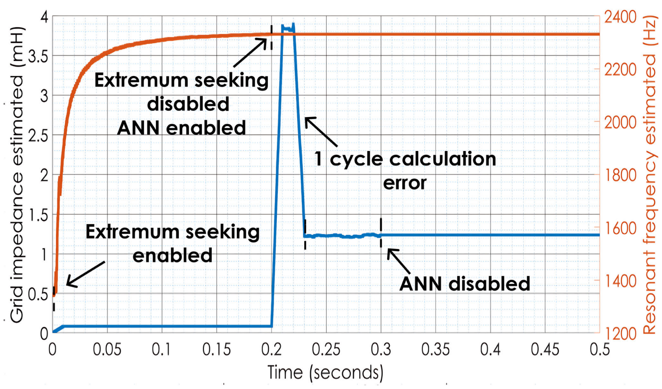

Fig. 10. Condition monitoring algorithm representing grid impedance and resonance estimation.

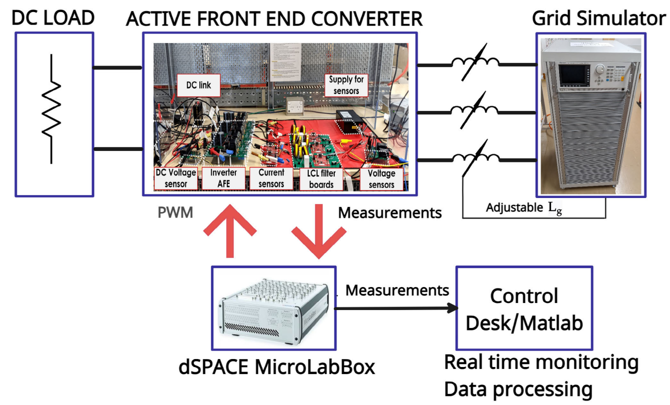

Fig. 11. Experimental setup consisting of active front-end converter, grid simulator, and dSpace MicroLabBox.

TABLE I
EQUIPMENT LIST

Function	Equipment
Voltage source	Chroma 61511 programmable AC source
Device under test	In-house Active front end converter
Controller	dSpace MicroLabBox
Load	Resistive
Software	dSpace control desk, Matlab

the ANN-based grid impedance estimation, showing accurate predictions across different grid conditions. Furthermore, there is potential to enhance the performance by training the ANN specifically on these diverse grid conditions.

B. Condition Monitoring Algorithm

Following ANN grid impedance estimation validation, the algorithm presented in Section III-D is tested on system, as seen in Fig. 2. The objective of this simulation is to estimate the capacitance of the filter capacitor. For this purpose, the values of the filter inductors on the grid side and converter side are known, while the capacitance of the capacitor and the grid impedance are unknown. In order to demonstrate the performance of the algorithm, an intentionally large and unrealistic initial guess is used for the capacitor capacitance. This is done to highlight the algorithm's ability to converge and accurately estimate the actual capacitance, even when starting from a significantly different initial value.

Figs. 9 and 10 illustrate the estimation of the capacitance in multiple segments, demonstrating the algorithm's performance throughout the simulation. Fig. 9 shows capacitor estimation in red, and Fig. 10 shows grid impedance and resonance estimation in orange and blue through the algorithm process. First, at initiation, ES is enabled for 0.2 s, and the estimation of resonant frequency starts. As seen in the figure, in both functions (capacitor estimation in red and resonance estimation in orange), the ES quickly converges to its local maximum. At this stage, the output of the resonance estimation is maintained and held at its converged value, and ES is deactivated. In addition, important to mention, the initial guess for the capacitor value (starts above 10 μF), intentionally set unrealistically wrong as evident from the figure, serves to showcase the algorithm's ability to detect substantial disparities between the assumed and actual values.

At the moment when ES is disabled, ANN is enabled. The figure displays an initial error occurring just after 0.2 s, attributable to the sliding window averaging over a one-cycle (fundamental frequency) in the magnitude extraction block. However, as soon as one cycle passes, the ANN outputs the accurate prediction of grid impedance. In order to mitigate the undesired noise introduced by the magnitude extraction process (as seen in the previous section), a moving average filter is applied to the estimated grid impedance before performing capacitor estimation. As seen from Fig. 10, the output of ANN oscillates around 1.2 mH, which is the real value of the grid impedance. At the start of 0.3 s, the ANN is disabled, and the last prediction of grid impedance is held. That is also the last calculation done by the algorithm, and the last value held is the capacitor estimation. As seen from the simulation, the estimated capacitance is around 2.99 μF, which is extremely close to the real capacitor value of 3 μF. A similar low error can be obtained for different capacitor values, LCL filters, and grid impedances. The magnitude of this error primarily depends on the quality of the ANN training, with a minor influence from the convergence error of ES. In summary, the entire process is highly efficient, taking only 0.3 s to complete. Therefore, the power quality pollution resulting from ES is not a concern within this short timeframe.

TABLE II
COMPARISON OF THE PROPOSED SOLUTION WITH THE EXISTING APPROACHES

Technique	Estimation speed	Extra sensor	LCL filter AC capacitor specific	Accuracy	Estimates or takes into account the impact of grid impedance
Proposed	Fast	No	Yes	Very good	Yes
AC filter MPPF capacitor estimation ([16])	Fast	No	Yes	Good only if there is no effect of grid impedance	No
Condition monitoring of DC link capacitors ([12])	Fast	Yes(3)	No	Very good	No

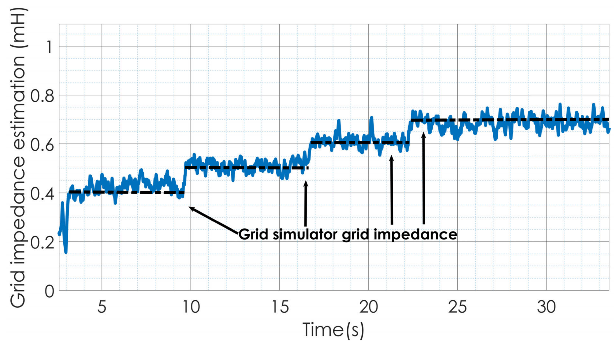

Fig. 12. Experimental validation of grid impedance estimation using ANN.

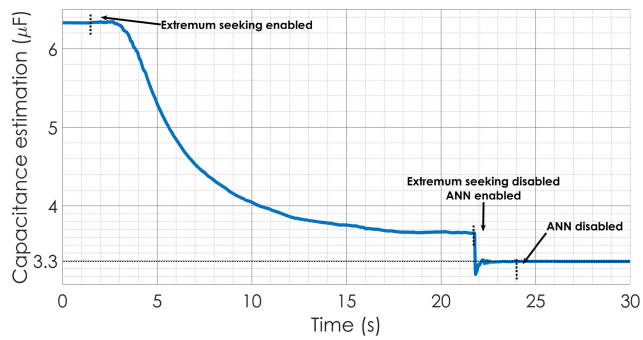

Fig. 13. Experimental results for condition monitoring algorithm capacitance estimation.

V. EXPERIMENTS

The experimental section provides a validation of the ANN grid impedance estimation and proposed condition monitoring algorithm. The experimental setup, depicted in Fig. 11, listed in Table I involves real-time data monitoring and extraction via control desk software. Subsequently, MATLAB is employed for enhanced data visualization. The grid impedance is manually adjustable using a grid simulator.

A. Experimental Validation of ANN Grid Impedance Estimation

Similar to the simulation section, it is crucial to verify the capability of the ANN in accurately estimating grid impedance. The experiments consist of three manual step changes in grid impedance in real time, starting from 0.4 to 0.7 mH. The ANN used in experiments is trained based on simulation data and on Australian-rated grid voltage, while in experimental validation, a lower voltage was used due to HAS requirement. As seen in Fig. 12, ANN estimation shown in blue follows grid impedance step changes closely. Initially, with a grid impedance of 0.4 mH, the ANN provides close estimates with only minor errors. Around the 10-s mark, a step change occurs, and the impedance is altered to 0.5 mH. Two additional step changes occur at approximately 16 and 23 s. Despite the noise in estimation resulting from Fourier series magnitude extraction, the ANN accurately predicts the grid impedance during each of the step changes. In fact, these results serve as validation of the ANN's capability to accurately estimate grid impedance under step changes. The following experiments will now concentrate on the entire condition-monitoring algorithm.

B. Experimental Validation of Condition Monitoring Algorithm

The following experiment is testing condition monitoring algorithms on a slower time scale. The LCL filter used in experiments differs from simulations. The converter L_1 and grid side inductors L_2 have the value of 2 mH, while the filter capacitor is 3.3 μF with 5% of tolerance. Since it is important to test capacitance estimation under grid impedance, grid simulator impedance is adjusted to 0.5 mH to mimic grid parasitic impedance.

For better visualization and due to real-time monitoring and manual enabling and disabling of each algorithm subpart (ES and ANN) in the control desk software, the proposed algorithm convergence is slowed down by reducing the integral component in ES. This ensures slower convergence and the ability to monitor the algorithm real time and adjust accordingly.

As shown in Fig. 13, capacitance estimation starts around 1.5 s when the system is in a steady state, and ES is enabled. As mentioned, extremum-seeking convergence is slowed down for real-time monitoring and visualization. Similar to the simulation, the initial guess or starting point is deliberately set far from the real capacitance value to demonstrate the accuracy of the estimation. ES converges around 16 s, but it remains active, to showcase exhibiting oscillations around the local extremum point even after convergence. At around 22 s, ES is disabled, the last value is held, and ANN is enabled. After one cycle calculation error, ANN instantaneously estimates grid impedance correctly, which can be seen in the figure, as the capacitance value is correctly estimated. Therefore, the algorithm successfully detects the capacitance of the filter capacitor.

It is noteworthy that both ES and ANN may introduce minor calculation errors under specific conditions; however, in this instance, these errors offset each other, resulting in highly accurate outcomes. In industry applications, the initial value of capacitance would be compared with measurements/estimations, then, even if small errors are present, that would not negatively affect the condition monitoring outcome, determining the end of the life of the filter capacitor. Thus, experimental results validate the algorithm's ability to monitor filter capacitor capacitance, which is a crucial factor in determining the degradation and life of the cycle of the capacitor.

VI. Conclusion

This article addresses the critical issue of ac MPPF filter capacitor lifespan in AFE applications and introduces an algorithm for capacitor capacitance detection, providing a seamless integration option for the existing systems.

The proposed algorithm relies on the idea that capacitor capacitance can be determined through LCL filter resonant frequency. However, to be able to do so, the grid impedance effect needs to be considered as well. For this reason, the algorithm is divided into two subparts: ES control, which detects system resonance, and an ANN that estimates grid impedance. Ultimately, the combination of these two subparts allows for the estimation of filter capacitance.

The algorithm and its two subparts underwent extensive investigation and validation. Simulations demonstrated the potential of ANN for grid impedance estimation, and the condition monitoring algorithm showed rapid and precise filter capacitor capacitance estimation. Experimental tests on a real AFE system validated these findings.

Compared with alternative methods, this technique is tailored specifically for application in ac capacitor LCL filters, thus offering inherent advantages over similar approaches. Table II shows the proposed solution in comparison with the existing approaches. Two specific publications have been selected for comparison. The proposed solution is specifically tailored for monitoring ac capacitors in LCL filters, similar to the approach described in [16]. In addition, the comparison includes an extremely accurate method for monitoring dc-link capacitors, as detailed in [12]. While the method described in [12] is not explicitly tailored for ac capacitors in LCL filters, it could potentially be adapted for such applications. However, this adaptation would likely require the addition of three extra sensors: capacitor current, voltage, and temperature sensors, representing a significant trade-off. In fact, the majority, if not all, of the techniques discussed in the introduction necessitate the use of at least one or more additional sensors primarily because of the underlying principles governing the calculation of capacitance or ESR. The proposed solution and the method described in [16] utilize only the existing current sensors in the LCL filter, harnessing the inherent LCL frequency to monitor changes in the ac capacitor. This makes them more practical for implementation in LCL filter converter applications. The primary drawback of [16] is its failure to consider the impact of grid impedance on the resonant frequency, raising doubts about its accuracy. In contrast, the method proposed in this article addresses this limitation by accounting for grid impedance, resulting in more accurate capacitance estimation. It is worth noting that the method described in [12] computes both ESR and capacitance, potentially providing more robust estimates compared with the proposed method and [16]. However, this advantage comes with the trade-off of requiring more computational power. In summary, while [12] provides accurate capacitor degradation estimation, it faces drawbacks for not being tailored for LCL filter ac capacitors, like the need for additional sensors and higher computational demand. On the other hand, [16] was specifically tailored for LCL filter ac capacitors, and however, fails to consider grid impedance effect LCL filter resonant frequency, which should affect the estimation. That is why the proposed method stands out by employing a unique approach to precisely monitor LCL filter ac capacitors without requiring additional sensors. In conclusion, the proposed algorithm, which synergistically combines the advantages of ES and ANN, offers a promising technique for ac film capacitor condition monitoring, potentially serving as a valuable monitoring tool in industrial applications.

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Edi Matijevic (Member, IEEE) received the M.Sc. degree in energy engineering with specialization in power electronics and drives engineering from Aalborg University, Aalborg, Denmark, in 2019, and the Ph.D. degree in electrical engineering from The University of Queensland, Brisbane, QLD, Australia, with focus on power electronics, in 2024.

He is currently a Research and Development Power Electronics Engineer with EcoJoule Energy, Brisbane. His research interests include power electronics converters design and control, electrical drives, and power quality.

Rahul Sharma (Senior Member, IEEE) received the M.Eng.Sc. and Ph.D. degrees in electrical engineering from The University of Melbourne, Melbourne, Australia, in 2004 and 2008, respectively.

He is currently an Associate Professor with the School of Electrical Engineering and Computer Science, The University of Queensland, Brisbane, QLD, Australia. He is the Lead Founder of $Solaris^{AI}$ Pty Ltd., Brisbane. His research interests include control theory development and applications to control of grid connected inverters, demand management algorithms, and monitoring of large solar farms.

Dr. Sharma was a recipient of SAE-A Young Engineer of the Year Award in 2010, and number of industry and academia selected best paper awards.

Ajay Hemanth is an electrical engineer and a data scientist. He is currently the Manager of AI and Data Analytics, The University of Queensland, Brisbane, QLD, Australia. Specializing in mathematical modeling, his research focuses on harnessing machine learning for predictive maintenance of solar farms and analyzing electrical harmonics.

Firuz Zare (Fellow, IEEE) was the Head of the School of Electrical Engineering and Robotics, Queensland University of Technology, Brisbane, QLD, Australia. He has over 20 years of experience in academia, industry, and international standardization committees, including eight years in two large research and development centers in Denmark, working on grid connected inverters, energy conversion systems, and power quality projects. He was an active member of IEC, Danish, and the Australian Standardization Committees. He was a Task Force Leader (International Project Manager) of Active Infeed Converters to develop the first international standard IEC 61000-3-16 within the IEC standardization SC77A. He has published over 300 peer-reviewed conference and journal papers, including four books and supervised over 20 Ph.D. students, postdocs, and research engineers. His research interests include advanced power converter topology and control in grid-connected renewable energy, motor drives and energy storage systems, electromagnetic interferences and harmonics in power systems, addressing standardization and emerging issues of future grids, and pulsed power systems for bioelectric and industrial applications.

Prof. Zare is a fellow of Engineers Australia, an IEEE Distinguished Lecturer. He received several awards, such as the Australian Future Fellowship, the John Madsen Medal, the Symposium Fellowship, and the Early Career Excellence Research Award. He is a Senior Editor of IEEE ACCESS journal, a Guest Editor and an Associate Editor of the IEEE JOURNAL OF EMERGING AND SELECTED TOPICS IN POWER ELECTRONICS, and an Editorial Board Member of several international journals.

Dinesh Kumar (Senior Member, IEEE) received the Master of Technology (M.Tech.) degree in power system engineering from Indian Institute of Technology (IIT) Roorkee, Roorkee, India, in 2004, and the Ph.D. degree in power electronics from the University of Nottingham, Nottingham, U.K., in 2010.

From 2004 to 2005, he held the position of a Lecturer with the Electrical Engineering Department, National Institute of Technology, Kurukshetra, India. In 2006, he joined the Technical University of Chemnitz, Chemnitz, Germany, as a Research Fellow of power electronics. Since 2011, he has been associated with Danfoss Drives A/S, Gråsten, Denmark, where he has been contributing to numerous research and industrial projects. He is an Adjunct Associate Professor at the School of Electrical Engineering and Robotics, Faculty of Engineering, Queensland University of Technology, Brisbane, QLD, Australia. His current research focuses on motor drives, harmonic analysis and mitigation techniques, power quality, and electromagnetic interference in power electronics.

Dr. Kumar was a recipient of two IEEE Best Paper Awards. He was appointed as a Distinguished Lecturer for the IEEE Industry Application Society from 2024 to 2025. He is a Technical Expert of Danish National Standards Committee in the IEC Standardization Working Group in TC77A, TC22/SC22G, and SyC LVDC. He is the Editor-in-Chief of the *International Journal of Power Electronics*. He serves as an Associate Editor Member for IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS and the Editorial Board Member of the IEEE Transportation Electrification eNewsletter.